The Neurological Bases of Human Behavior

Fall 2026 ▪ MWF 1:25-2:15 pm

2026-07-14

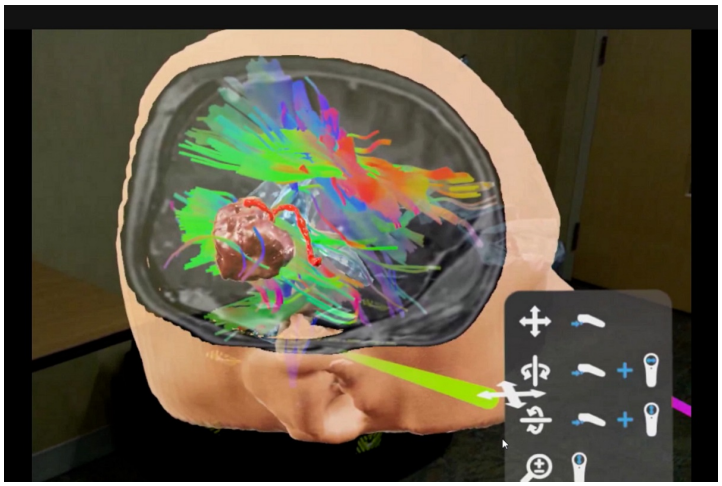


Figure 1: "A GPS for your mind" (2023)

Themes

There is no more fascinating or challenging field of inquiry than the nature, origin, structure, and functioning of the human mind and the behavior it enables. This course will examine human mental states and observable behaviors from a biological basis, especially focusing on the evolution, development, structure, and functioning of the nervous system.

Week 01

Mon, Aug 24

Course Introduction

- *Slides*: | html |

Wed, Aug 26

History of neuroscience, levels of analysis, methods

- *Read*: https://en.wikipedia.org/wiki/History_of_neuroscience, https://en.wikipedia.org/wiki/Level_of_analysis
- *Watch*: National Geographic (2014), NeuroBriefs (2011), Eames Office (2010)
- *Slides*: | html |

Fri, Aug 28

Methods

- *Explore*: https://en.wikipedia.org/wiki/List_of_neurological_research_methods
- *Watch*:
  - National Institute of Biomedical Imaging and Bioengineering (2013)
  - National Institute on Biomedical Imaging and Bioengineering (n.d.) - Nature Video (2010)
  - Understanding Animal Research (2009)
  - Video, Nature (2013)
- *Slides*: | html |

Week 02

Mon, Aug 31

Neuroanatomy I

- *Read*: https://en.wikipedia.org/wiki/Neuroanatomy
- *Slides*: | html |

Wed, Sep 2

Neuroanatomy II

- *Slides*: | html |

Fri, Sep 04

Neuroanatomy III

- *Slides*: | html |

Week 03

Mon, Sep 07

NO CLASS: Labor Day

Wed, Sep 09

Quiz 1; Cells of the nervous system

- *Read*: https://en.wikipedia.org/wiki/Neuron, https://en.wikipedia.org/wiki/Glia
- *Slides*: | html |

Fri, Sep 11

Neurophysiology I

- *Read*: https://en.wikipedia.org/wiki/Resting_potential
- *Slides*: | html |

Week 04

Mon, Sep 14

Neurophysiology II

- *Watch*: khanacademymedicine (2012)
- *Slides*: | html |

Wed, Sep 16

Neurophysiology III

- *Read*: https://en.wikipedia.org/wiki/Action_potential
- *Slides*: | html |

Fri, Sep 18

Catch-up; Exam 1 review

- *Study Guide*: | html |

Week 05

Mon, Sep 21

Exam 1

Wed, Sep 23

Neurochemistry I

- *Read*: https://en.wikipedia.org/wiki/Neurotransmitter
- *Slides*: | html |

Fri, Sep 25

Neurochemistry II

- *Read*: https://en.wikipedia.org/wiki/Glutamic_acid, https://en.wikipedia.org/wiki/GABA, https://en.wikipedia.org/wiki/Acetylcholine, https://en.wikipedia.org/wiki/Monoamine_neurotransmitter
- *Slides*: | html |

Week 06

Mon, Sep 28

Hormones

- *Slides*: | html |

Wed, Sep 30

Evolution

- *Read*: https://en.wikipedia.org/wiki/Evolution_of_the_brain
- *Watch*: Scale: (2023), The Economist (2015)
- *Slides*: | html |

Fri, Oct 02

Development

- *Skim*: https://en.wikipedia.org/wiki/Development_of_the_nervous_system
- *Slides*: | html |

Week 07

Mon, Oct 05

Quiz 2; Exam 1 results

- *Slides*: | html |

Wed, Oct 07

Psychopathology I

- *Review*: https://en.wikipedia.org/wiki/Major_depressive_disorder
- *Slides*: | html |

Fri, Oct 09

Psychopathology II

- *Review*: https://en.wikipedia.org/wiki/Bipolar_disorder
- *Slides*: | html |

Week 08

Mon, Oct 12

Psychopathology III

- *Review*: https://en.wikipedia.org/wiki/Schizophrenia
- *Slides*: | html |

Wed, Oct 14

Emotion I

- *Slides*: | html |

Fri, Oct 16

Emotion II

- *Slides*: | html |

Week 09

Mon, Oct 19

Catch up; Exam 2 review

- *Slides*: | html |

Wed, Oct 21

Exam 2

Fri, Oct 23

Emotion III

- *Slides*: | html |

Week 10

Mon, Oct 26

Sleep I

- *Slides*: | html |

Wed, Oct 28

Sleep II

- *Slides*: | html |

Fri, Oct 30

Sleep III

- *Slides*: | html |

Week 11

Mon, Nov 02

Vision I

- *Slides*: | html |

Wed, Nov 04

Vision II

- *Slides*: | html |

Fri, Nov 06

Quiz 3; Exam 2 results

Week 12

Mon, Nov 09

Vision III

- *Slides*: | html |

Wed, Nov 11

Audition

- *Slides*: | html |

Fri, Nov 13

Sensory Systems

- *Slides*: | html |

Week 13

Mon, Nov 16

Somatosensation

- *Slides*: | html |

Wed, Nov 18

Action I

- *Slides*: | html |

Fri, Nov 20

Quiz 4; Action II

- *Slides*: | html |

Thanksgiving Break

NO CLASS

Week 14

Mon, Nov 30

Exam 3 review

Wed, Dec 02

Exam 3

Fri, Dec 04

Cognition

Week 15

Mon, Dec 07

Memory I

- *Slides*: | html |

Wed, Dec 09

Memory II; Exam 3 results

Fri, Dec 11

Exam 4 (final) review; The Cerebral Symphony

- *Slides*: | html |

Finals Week

Exam 4 (Final) TBD

References

A GPS for your mind: Penn State Health's Brainlab surgical system helps teen recover from hemorrhage. (2023, October 30). Retrieved August 18, 2025, from https://pennstatehealthnews.org/2023/10/a-gps-for-your-mind-penn-state-healths-brainlab-surgical-system-helps-teen-recover-from-hemorrhage/

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